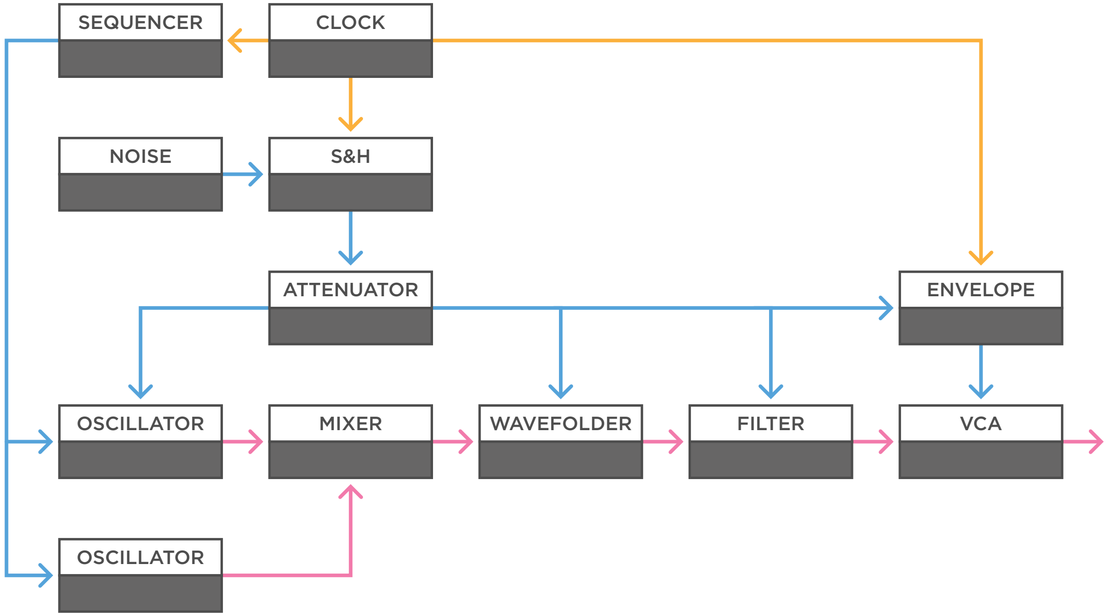
# Sample & Hold

## Stepped Random

Audio
Control Voltage
Trigger / Gate

MONOTRAIL TECH TALK




(a)

a - If you take a sample & hold module, feed it noise, and use a clock to trigger it, you create a classic stepped random voltage that can be used to modulate parameters. If you make a voice with two oscillators, a mixer, wavefolder, filter, and a VCA, you can experiment with sending the sample & hold signal to different elements in the voice.

If you add a sequencer and a voltage-controlled envelope, you can also control the attack or decay time with the stepped random voltage, to create dynamic sequences. Quite often though, attenuating the signal is the key to good sounding results.

# Sample & Hold

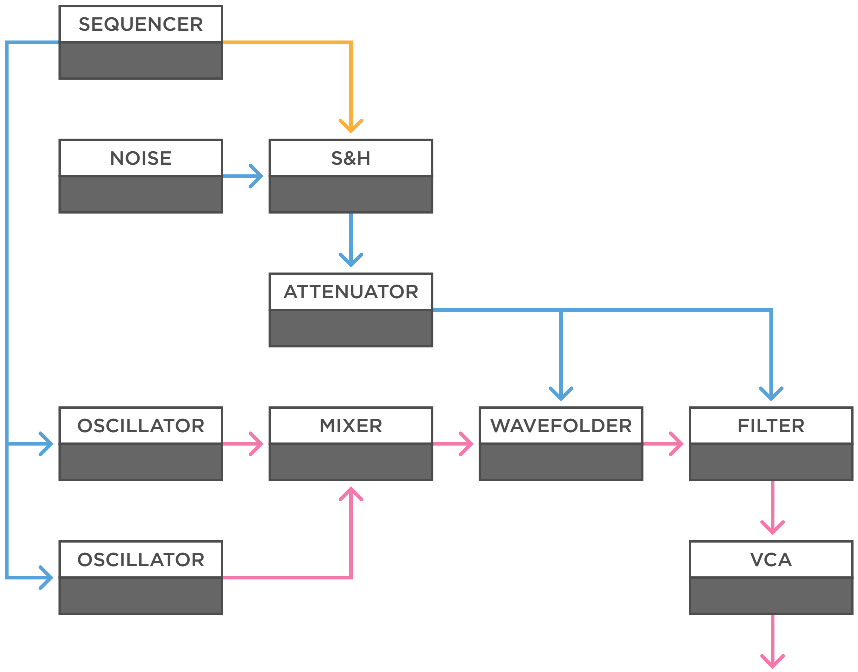
## Stepped Random

MONOTRAIL
TECH TALK

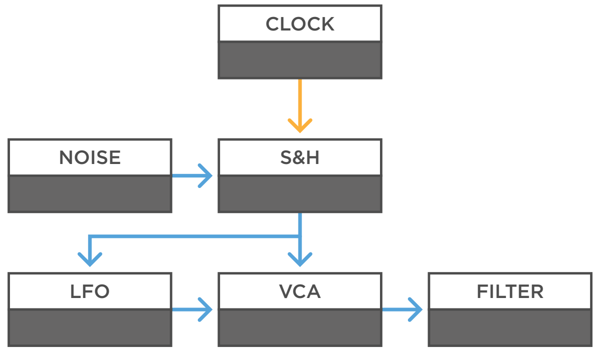
Audio

Control Voltage

Trigger / Gate




(a)




(b)

a - A nice aspect of the sample & hold signal is that it's random, but steady between triggers. If you manually play the voice, for example with a sequencer like the Beatstep Pro, you can use its gate out to the trigger input of the sample & hold module. Now, every time you play a note, the gate triggers the sample & hold module, which generates a new steady value.

b - Another way to creating interesting dynamics is to use the random sample & hold signal on a VCA and use that VCA to control the amount of modulation between two modules. For example, the amount of frequency modulation between two oscillators, or the amount of modulation from an LFO to a filter. In the latter case you can multiply the sample & hold signal to control the speed of the LFO as well.

# Sample & Hold

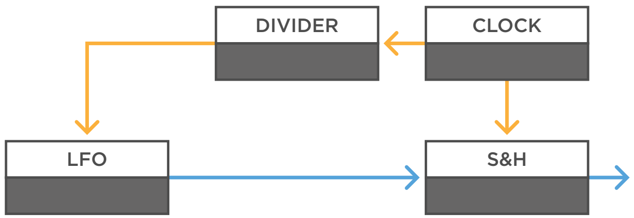
## Input Variations

Audio

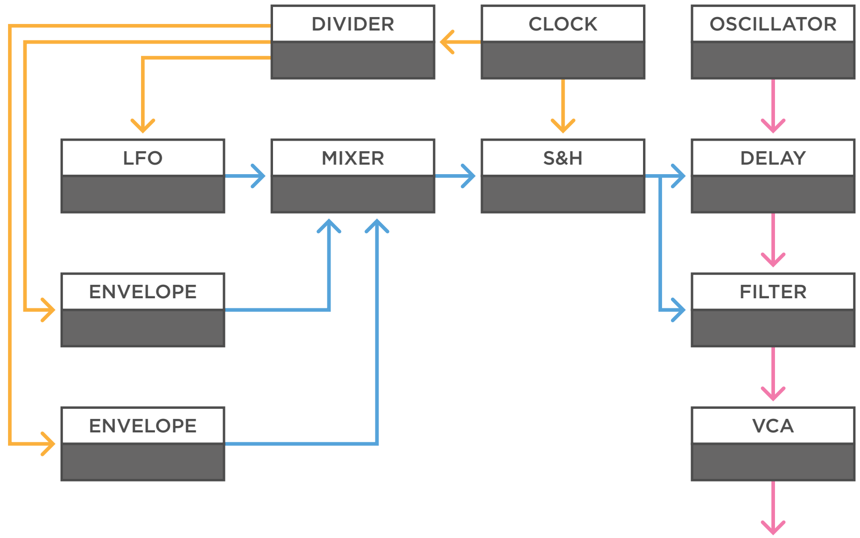
Control Voltage

Trigger / Gate






(a)




(b)

a - If you use noise as the input signal with a steady clock, you get a stepped random signal. But you can feed the sample & hold anything you like. LFOs can be fun as well. You can use free running LFOs to create shifting patterns. However, if you want the pattern to repeat itself, you can send the same clock you use to trigger the sample & hold, to a clock divider, and have a division of the clock re-trigger the LFO. This way the clock and LFO shape are exactly in sync, and you will create a looping sample & hold pattern.

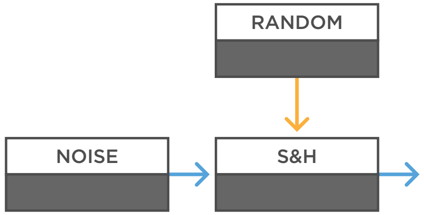
b - You can make the pattern as complex as you like. For example, you can use a few different clock divisions to re-trigger an LFO, and gate two envelopes. Then you can use a mixer to mix these signals together and feed the result to the sample & hold. You can create longer looping patterns depending on what divisions you use. In this example the sample & hold signal is used on a simple voice and sent to the filter, as well as the delay time, to create a more industrial rhythmical patch. By mixing in the different signals by hand you can mix in different looping patterns.

# Sample & Hold

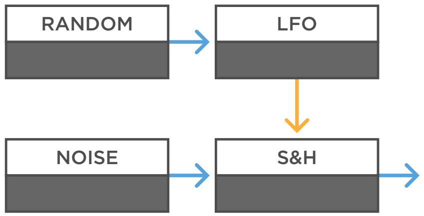
## Trigger Variations

MONOTRAIL
TECH TALK

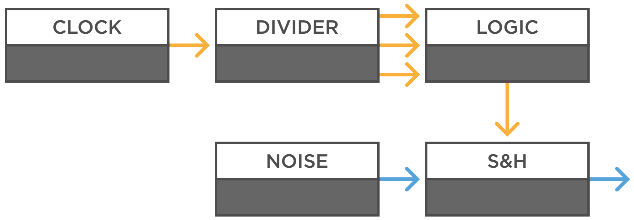
Audio
Control Voltage
Trigger / Gate



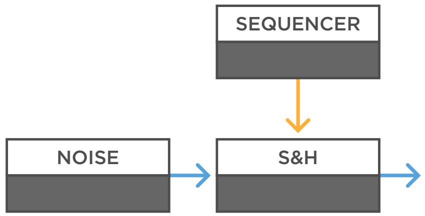

(a)




(b)




(c)




(d)

a - You can increase the randomness of a sample & hold signal by using an un-synced random trigger signal, instead of a steady clock. This way you get random voltages, generated at random moments.

b - You can experiment with the trigger input signal to create different patterns. For example, you can use a simple square wave LFO, but send the LFO a smooth random voltage to make its speed drift over time.

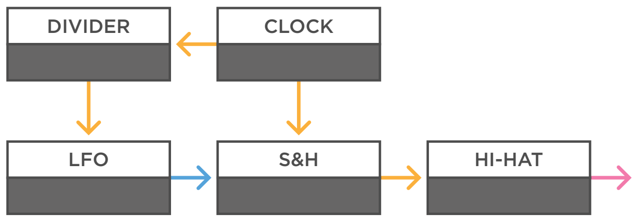
c - Or you can use a combination of a clock, divider, and logic module, to create tempo synced but interesting changing patterns.

d - Or use a simple trigger sequencer to create a custom pattern.

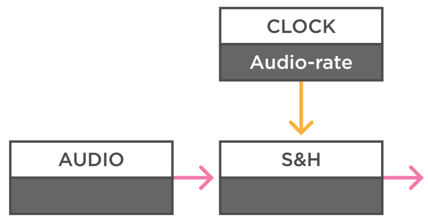
# Sample & Hold
## Triggers / Audio



Audio
Control Voltage
Trigger / Gate




(a)




(b)

a - A simple way to create interesting percussive triggers is to feed an un-synced square wave into the sample input, and a fast clock to the trigger input of the sample & hold module. By changing the frequency of the LFO you can create interesting patterns. And if you want, you can create looping patterns by using a clock divider and re-triggering the LFO module every few bars.

b - If you have an analog sample & hold module, you can use it as a rudimentary sample rate reducer. When you feed an audio signal into the sample input, the trigger input will determine the sample rate. And if you want to get audible sounds out of this, you need to feed a high audio rate trigger signal into the trigger input.